

Answers written in the margins will not be marked.

*2. A gas bubble rises from the bottom of a lake to the water surface. Its radius increases from 0.8 cm to 1.0 cm.

- (a) If the gas pressure in the bubble at the water surface is 1.01×10^5 Pa, find the gas pressure in the bubble when it is at the bottom of the lake. Assume that the temperature of the gas in the bubble remains constant. (2 marks)

- (b) Use kinetic theory to explain the change in gas pressure in the bubble as its volume increases. (2 marks)

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- (ii) If this falling person of mass 70 kg is stopped in 0.3 s by the life net, estimate the average force acting on the person by the net within this time interval. (3 marks)

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- (iii) What form of energy is stored by the life net during the deceleration of the falling person ? (1 mark)

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- (b) (i) Give a reason why there exists a height limit of using life nets. (1 mark)

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- *(ii) The falling person might hit the rim of the net, thus the person or the firemen holding the rim would be injured. Explain why it is not easy for a person jumping from a height to reach the life net's central part. (2 marks)

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- (ii) The total mass of the cyclist and the bicycle is 65 kg. Calculate the resistive force for the deceleration. (2 marks)

- (iii) Find the distance between the bicycle and the obstacle when the bicycle just stops. (3 marks)

- (d) Figure 3.2 shows a knee pad with a padded cushion on the front to protect the cyclist in the event the knee hits the ground. Explain the working principle of this padded cushion. (2 marks)

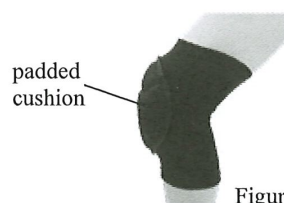


Figure 3.2

Answers written in the margins will not be marked.

- *4. (a) Figure 4.1(a) shows a hollow cylindrical drum rotating uniformly about its central vertical axis. With a high enough rotation speed, the block shown in the figure appears 'pinned' to the wall of the drum without falling.

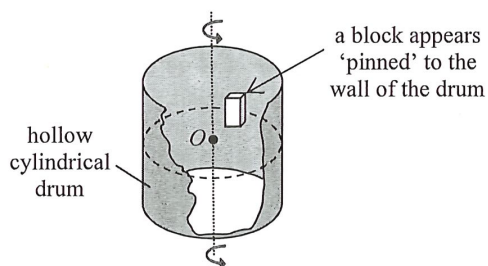


Figure 4.1 (a)

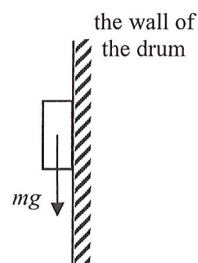


Figure 4.1 (b)

- (i) Besides the block's own weight mg , there are two more forces acting on it. Indicate and label these two forces on Figure 4.1(b). (2 marks)
- (ii) Which force provides the centripetal force for the block ? (1 mark)

- (b) A washing machine has a hollow cylindrical drum with numerous small holes that allow water droplets to escape. In 'spin' mode, the drum's angular speed ω is 78.5 rad s^{-1} .

- (i) What is the rotation speed of the drum in revolutions per minute ? (2 marks)

The cylindrical drum of the washing machine is of radius 0.25 m . Figure 4.2 shows the top view of the drum spinning anticlockwise about its central vertical axis through O .

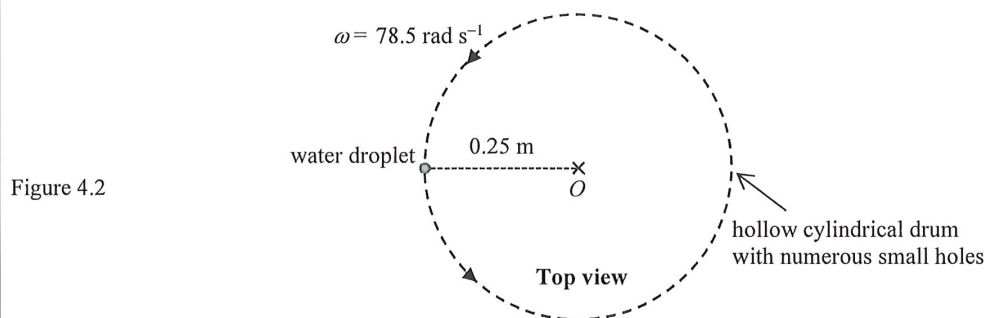
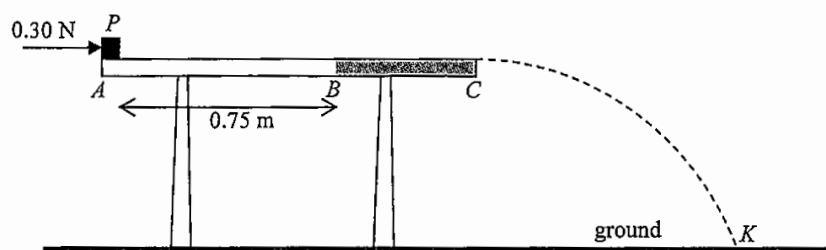


Figure 4.2

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3. A small block P of mass 0.20 kg rests on side A of a horizontal table as shown. The surface of the table is smooth from A to B but is rough from B to C . Neglect air resistance.

Figure 3.1



A horizontal force of 0.30 N acts on the block as shown.

- (a) Find the speed of the block after it travels 0.75 m to point B . (2 marks)

- (b) The horizontal force is removed when block P just reaches B . When the block continues to travel from B to C , on the free-body diagram below indicate and name the force(s) acting on the block. (2 marks)



- *(c) The block leaves side C of the table and takes 0.35 s to reach point K on the ground. Estimate the height of the table. (2 marks)

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